

TECHNICAL WHITE PAPER

# CourtRank

A Transparent Ranking System for Rotating-Partner Doubles Pickleball  
Ladders

*Design rationale, algorithm specification, and Monte-Carlo validation*

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## Revision History

| Version | Date         | Change                                                                                                                                                                                                                                                                   |
|---------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | June 2026    | Initial release: algorithm, simulation evidence, operating guide.                                                                                                                                                                                                        |
| 1.1     | June 2026    | Full mathematics as numbered display equations; notation appendix.                                                                                                                                                                                                       |
| 2.0     | Jun 21, 2026 | Weekly batch scoring adopted before launch: all of a session's games scored against the week-start state (ADR 0001).                                                                                                                                                     |
|         | Jul 10, 2026 | Engine's movement-clamp default permanently disabled and rating floor removed (ADR 0002–0003); the paper's ranking description was already cap-free.                                                                                                                     |
|         | Jul 17, 2026 | Seed correction tapered to zero over 12 games and blend reduced 0.6 → 0.1 (ADR 0004–0005; commit f536c62).                                                                                                                                                               |
|         | Jul 2026     | Paper rewritten to the deployed configuration: simulation results and worked examples regenerated; severe mis-seeds now correct conservatively by design, with organizer reseeding the intended remedy for implausible seeds. Verified against repository state 5a00c2e. |

## Executive Summary

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CourtRank ranks an eight-week, rotating-partner doubles pickleball ladder of roughly thirty-six players across six courts. Each player's DUPR seeds a starting rating; weekly results adjust it, and courts are dealt by a straight sort of ratings. A session's games are evaluated together, as a batch, against the ratings that opened the week, so the outcome never depends on score-entry order.

The engine is margin-aware Elo with two additions for this format. A contribution split addresses the central difficulty of doubles — how much credit each partner earns from a shared result — by pairing a per-game heuristic with the natural experiment of partner rotation. A small, gated seed correction nudges shadow-graded newcomers toward their demonstrated level during their first twelve games, tapering to zero over that window; afterward every player advances through ordinary results alone.

Across 150 simulated seasons per scenario at the deployed configuration, ranking accuracy reaches  $p = 0.91$  by week eight, 93% of players finish within one court of their true skill, rating inflation is negligible, and — with no cap on movement — at least 99% of player-weeks move six rank positions or less. The design deliberately favors accurate ordinary-season ordering and transparent movement over aggressive automatic repair of extreme administrative mis-seeds: a grossly implausible seed is a data problem for the organizer, not a case the algorithm must overpower. The rest of the paper documents the reasoning, the mechanics, and the evidence.

## 1. The Problem and Its Constraints

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The league presents a deceptively simple request — rank the players each week — wrapped around constraints that interact in non-obvious ways. Three of them shaped every subsequent design decision.

### 1.1 Attribution in doubles is unidentifiable within a single game

From the final score of one doubles game, there is no statistical way to recover how much each partner contributed. The score is a single observation of a two-person system; any split between partners is an assumption, not a measurement.

The rotating-partner format, however, is a designed experiment. Each player partners several different people and faces shifting opposition, which makes each player the one common factor across their own four-plus games — and a player who wins regardless of partner reveals their strength through that consistency. The system therefore uses a defensible per-game heuristic and lets the rotation perform the real attribution across the session. In testing, a hidden strong player seeded identically to weaker peers separated from them by roughly 113 rating points after a single session, correctly identified in 84% of runs.

### 1.2 Stability and responsiveness

A ladder must be responsive enough that a mis-ranked player reaches their true court within a few weeks, yet stable enough that one lucky night does not send someone rocketing across the courts. These goals pull in opposite directions. CourtRank reconciles them not with a hard cap on movement but through the update size itself: a settled player's rating shifts only modestly in any single week, so their court shifts modestly too. Ranking is then a plain sort by rating, which keeps a player's court and their points always in agreement — a player whose rating is sixth-best is in court one, with no exceptions to explain.

That stability is empirical — created by bounded rating updates, not enforced by rank manipulation — and the evidence bears it out: at the deployed configuration, at least 99.1% of simulated player-weeks moved

no more than six rank positions, and moves beyond twelve occurred in at most 0.007%. In live week-one data, 32 of 34 players stayed within roughly one court; the largest move, a 20-position climb, was a mis-seed correcting itself. There is deliberately no formal upper bound, so a rare large correction remains possible when the evidence demands one.

### 1.3 A constraint from the rotation itself

With six players on a court, it is impossible for everyone to partner everyone exactly once: that would need fifteen pairings at two per game, or seven and a half games. The practical rotation is six games in which each player plays four, with four distinct partners. A five-player court is perfect — five games, every pair partnered once — which is why courts of five are used when attendance is not a multiple of six.

## 2. The CourtRank Algorithm

CourtRank is a margin-aware, contribution-weighted team Elo with one safety valve. It has six working parts. Notation is introduced where it first appears and collected in Appendix B, the notation of record.

### 2.1 Seeding

Every player's DUPR — real where available, an organizer-assigned shadow grade otherwise — is converted to a rating on a 1500-centered scale. Because the conversion is a straight line, sorting by seed rating gives the same order as sorting by DUPR: week-one courts are just the players listed highest to lowest and dealt six at a time.

$$R_0 = 1500 + (\text{DUPR} - 3.5) \times 200 \quad (1)$$

No minimum rating is imposed; the scale is unbounded in both directions, at seeding and ever after. Verified and shadow grades use the same conversion, the same placement, and the same enlarged early update factor (Section 2.5). The tag changes exactly one thing: eligibility for the gated seed correction.

### 2.2 The weekly batch

Every game in a session is evaluated against the state that opened the week. At session start the engine freezes each player's rating, games-played count, update factor, and contribution share; each game's update is computed against that frozen snapshot, the per-player deltas are accumulated, and the sum is applied once:

$$R_i^{\text{after}} = R_i^{\text{start}} + \sum_{g \in \text{session}} \Delta R_{i,g} \quad (2)$$

Games-played counts also advance only after the batch is scored. A session is conceptually simultaneous — spreadsheet row order should not decide which games face changed ratings — so the batch rule buys input-order independence, reproducibility, and easy auditing: shuffling the game list changes nothing. Against the older sequential rule, headline accuracy moved by less than 0.001 in rank correlation. Determinism came free.

### 2.3 Per-game rating updates with margin

Each game is scored on its own merits. A team's rating is the average of its two partners,

$$R_{\text{team}} = \frac{R_i + R_j}{2} \quad (3)$$

and the Elo expectation gives the probability that team A should beat team B given the rating gap between them:

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}} \quad (4)$$

The actual result blends a win/loss term with a point-margin term:

$$S = \beta W + (1 - \beta) \left[ 0.5 + 0.5 \cdot \frac{s_f - s_a}{11} \right] \quad (5)$$

where  $W$  is the win indicator (1 for a win, 0 for a loss),  $s_f$  and  $s_a$  the points scored for and against, and  $\beta = 0.5$  weights the two terms equally.

The team's rating change is then the usual Elo product ( $K$  the update factor, Section 2.5):

$$\Delta_{\text{team}} = K (S - E) \quad (6)$$

The swing is symmetric between the teams, and a game among four settled players sharing the steady-state update factor is exactly zero-sum. Player-specific update factors, absence multipliers, and the winner floor (Section 2.4) introduce small deviations — measured drift is about +0.045 points per player per eight-week season.

A decisive 11-2 win moves ratings roughly twice as far as an 11-9 result — and the blend cuts both ways: an underdog who loses 9-11 when expected to lose badly has outperformed expectation, and in sufficiently extreme cases can gain rating while losing. The honest statement of the incentive is therefore not “win and you rise; lose and you fall” but: a player rises by outperforming the rating-and-margin expectation and falls by underperforming it, with the added guarantee that winning a game never produces a negative adjustment. Because games are scored against the ratings actually on the floor, the system is indifferent to which court a game is played on — players who cross between courts within a session need no special handling.

## 2.4 The contribution split

Within a team, a result is not divided evenly. Each partner's share of the swing is proportional to their rating, clipped so that no player ever absorbs more than 65% or less than 35%:

$$w_i = \text{clip}\left(\frac{R_i}{R_i + R_j}, 0.35, 0.65\right) \quad (7)$$

After the two shares are renormalized to sum to one, player  $i$ 's rating change is the per-player Elo product, with the sign mirrored for the losing team:

$$\Delta_i = K_i m_i (S - E) \cdot 2 w_i \quad (8)$$

where  $K_i$  and  $m_i$  are player  $i$ 's own update factor and absence multiplier (equations 10-11) — so the higher-rated partner, the presumed carrier, gains more from a win and loses more from a loss. The clip keeps the per-game guess humble; durable attribution comes from rotation, as a player who lifts different

partners accumulates credit no single game could justify. The split is a heuristic, not proof of who carried a given game.

One guardrail applies at the individual level: a winning player's raw adjustment is floored at zero,

$$\Delta_i^{\text{winner}} = \max(0, \Delta_i) \quad (9)$$

so a heavy favorite who wins narrowly is never punished for the margin.

## 2.5 Adaptive update size and the gated seed correction

New players use a larger update factor that decays toward the steady-state value over their first few games:

$$K(g) = K_\infty + (K_0 - K_\infty) e^{-g/\tau_K} \quad (10)$$

where  $g$  is games played,  $K_0 = 56$  at debut,  $K_\infty = 32$  at steady state, and  $\tau_K = 10$  games sets the decay.

A returning absentee receives a temporary multiplier on top of this, capped so it cannot run away — a lightweight stand-in for formal uncertainty tracking:

$$m = \min(1.5, 1 + 0.15a) \quad (11)$$

where  $a$  counts consecutive missed weeks.

The most carefully tuned component handles shadow-graded players who sweep a session — win every game or lose every game. The correction fires only when all of the following hold:

- the player carries a shadow grade (verified-DUPR players are never subject to it);
- they entered the session with fewer than twelve prior games;
- the session contained at least three of their games; and
- they won every one, or lost every one.

For a correctly seeded player a sweep is a fluke — about 12.5% over four balanced games (either direction), 6.3% over five, 3.1% over six, 1.6% over seven — while for a gross mis-seed it is near-certain, which makes it a usable trigger. When it fires, the system computes the session performance rating  $P$  — the rating that would have made the day unsurprising:

$$P = 2(\langle O \rangle + L) - \langle T \rangle \quad (12)$$

where  $\langle O \rangle$  is the session-mean opponent-team rating,  $\langle T \rangle$  the session-mean partner rating, and  $L$  the session's Elo margin — the number of rating points by which the day's results outperformed an even match:

$$L = 400 \log_{10} \frac{\langle S \rangle}{1 - \langle S \rangle} \quad (13)$$

where  $\langle S \rangle$  is the session-mean team score, clamped to  $[0.08, 0.92]$  so the logit stays finite — capping  $L$  at about  $\pm 424$  points.

The rating is then pulled toward  $P$  by a strength that decays with experience:

$$\lambda = b e^{-g_p/\tau_p} \cdot m, \quad b = 0.1 \quad (14)$$

where  $b = 0.1$  is the blend ceiling,  $g_p$  the games played entering the session,  $\tau_p = 16$  games the decay constant, and  $m$  the absence multiplier of equation 11.

A dead zone  $z = 150$  suppresses the pull for gaps small enough to be session noise, the clip  $c$  shrinks linearly with experience, and eligibility ends outright at twelve games — roughly two real sessions at the league's six-to-seven games a week:

$$c(g_p) = 400 \cdot \max\left(0, 1 - \frac{g_p}{12}\right) \quad (15)$$

$$R \leftarrow R + \lambda \cdot \text{clip}(\text{sgn}(P - R) \cdot \max(0, |P - R| - z), -c, c) \quad (16)$$

The taper's meaning is deliberate: uncertainty belongs to the seed, not permanently to the player. After roughly two sessions the games have supplied the evidence and the mechanism disappears — a modest mis-seed nudge, not an alternate rating engine. Its ceiling:

| Games entering session | Clip $c$ | Blend $\lambda$ | Maximum pull (points) |
|------------------------|----------|-----------------|-----------------------|
| 0                      | 400      | 0.1             | 40                    |
| 4                      | 266.7    | 0.0779          | 20.77                 |
| 6                      | 200      | 0.0687          | 13.75                 |
| 7                      | 166.7    | 0.0646          | 10.76                 |
| 8                      | 133.3    | 0.0607          | 8.09                  |
| 11                     | 33.3     | 0.0503          | 1.68                  |
| 12                     | 0        | ineligible      | 0                     |

Table 1. Ceiling on the seed correction by experience (no absence multiplier). A clip-saturating debut sweep moves a rating by at most 40 points; a realistic two-session maximum is roughly 51–54 points.

## 2.6 Weekly re-rank

After each session every enrolled player is re-sorted by updated rating, and courts are dealt from that order — top six to court one, next six to court two, and so on. Ranking is a plain sort, so a player's court always matches their points exactly, and no artificial cap constrains how far a player may move; Section 4 quantifies how modest weekly movement stays on its own.

An absent player's rating is frozen; their ordinal position is not — active players may pass them, and the public ladder may list active players only. Mid-season joiners enter at the rank implied by their seed and are treated as uncertain for their first sessions; everyone's season improvement, originals and joiners alike, is measured from seed points and seed rank.

## 3. Why Not an Off-the-Shelf System

Glicko-2 and OpenSkill model player uncertainty rigorously, but they are opaque to participants and more machinery than thirty-six players with a clean weekly structure require. Their best idea — let less-certain ratings move faster — is borrowed here through the adaptive update size and the uncertain-player flag. DUPR itself is proprietary and not built for week-to-week ladder placement. Plain Elo is the right backbone because every number it produces can be reconstructed in a spreadsheet — which matters the first time a player disputes a court assignment.

## 4. Simulation and Validation

The system was tested against a simulator in which thirty-six players hold hidden true skills, receive noisy real or shadow seeds, play games rally by rally, and vary in weekly form. Four scenarios were each run for 150 seasons at the deployed configuration of Appendix A — weekly batching, no rank clamp, no floor, and the tapered 0.1-blend correction.

Rank correlation is reported as  $\rho$ , the Spearman correlation between true skill and assigned rating:  $\rho = 1.0$  means the ordering is perfect,  $\rho = 0$  no better than a shuffle — so  $\rho = 0.90$  means the rankings are very nearly in true order, off only in the occasional close pair. It measures order, not rating values, which is what matters for dealing courts.

| Scenario                           | Wk 1 $\rho$ | Wk 4 $\rho$ | Wk 8 $\rho$ | Exact court | $\pm 1$ court |
|------------------------------------|-------------|-------------|-------------|-------------|---------------|
| A. Baseline (full attendance)      | 0.856       | 0.889       | 0.913       | 54.8%       | 93.0%         |
| B. 12% weekly absences             | 0.851       | 0.882       | 0.905       | 55.0%       | 92.6%         |
| C. Absences + two wk-3 joiners     | 0.851       | 0.857       | 0.895       | 52.3%       | 90.7%         |
| D. Sandbagger (top-3, seeded last) | 0.726       | 0.786       | 0.842       | 50.1%       | 88.8%         |

Table 2. Core results across four stress scenarios, 150 simulated seasons each, at the production configuration. Court columns are measured at week 8.

By season's end, 93% of players sit within one court of their true skill; absences and mid-season arrivals barely move that. Inflation under the winner floor measured +0.045 points per player per season. And with nothing capping movement, it stays small on its own: at least 99.1% of player-weeks moved six positions — one court — or less, moves beyond twelve occurred in at most 0.007%, and the largest upward move in 600 simulated seasons was 12–13 positions. Larger real-world moves remain possible; when they occur they are essentially always a mis-seed correcting.

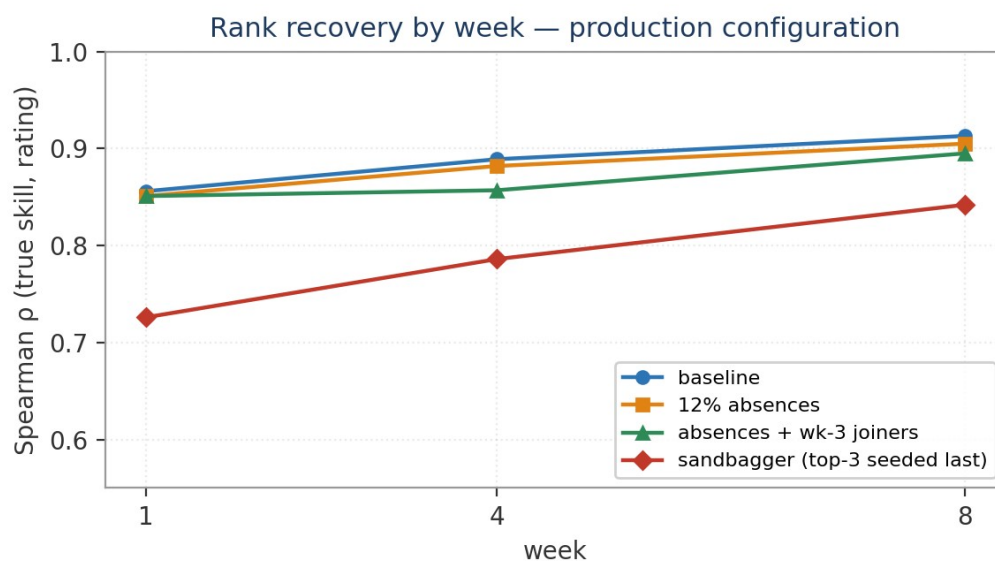


Figure 1. Rank recovery improves week over week and is robust to absences and joiners; even the severe mis-seed scenario recovers steadily.  $\rho$  is measured at weeks 1, 4, and 8.

## 4.1 Calibrating the seed correction

The pull blend  $b$  — how strongly a qualifying sweep drags a rating toward its session performance — was selected by sweeping  $b$  over  $\{0, 0.1, 0.2, 0.3, 0.6\}$  across 2,250 simulated seasons. The original 0.6 overreacted to single-session noise: a rec-league sweep still reflects partner draw, opponent draw, form, and a small sample, and letting one night move a debut rating by up to 240 points made real corrections violent and ordinary-season ordering measurably worse.

| Measure                            | Blend 0.6 | Blend 0.1 |
|------------------------------------|-----------|-----------|
| Baseline Week 8 p                  | 0.903     | 0.913     |
| Exact court, Week 8                | 54.3%     | 54.8%     |
| Within one court, Week 8           | 92.2%     | 93.0%     |
| Severe mis-seed, Week 8 court      | 3.54      | 4.61      |
| 300-point-low joiner, Week 8 court | 2.22      | 2.53      |

Table 3. The blend decision, evaluated with the 12-game taper active in both columns. Lower court numbers are better in the last two rows.

Suppressing the noise improved every ordinary measure. The cost is deliberate and visible in the last two rows: extreme mis-seeds now correct more conservatively, a trade examined next.

## 4.2 The hardest case: severe mis-seeds

The most demanding test drove each generation of the design. A strong player seeded dead last climbs too slowly under partner-diluted Elo alone; the gated correction exists to rescue exactly this case, restricted to shadow grades so the well-seeded majority is never destabilized. What changed in this revision is how much rescue it is allowed to perform.

Under the deployed configuration, a player seeded roughly 700 points low reaches about court 4.6 of six by week eight; a realistic 300-point-low mid-season joiner reaches about court 2.5 (Figure 2). The earlier, stronger correction delivered such a player to the upper courts by mid-season; this version does not — an intentional safety choice. Ordinary seed error self-corrects within the season. Extraordinary seed error proceeds conservatively, because the algorithm refuses to let one or two sessions overpower all prior placement evidence; a grossly implausible seed is an administrative mistake, and the remedy is administrative — the organizer reseeds (Section 5).

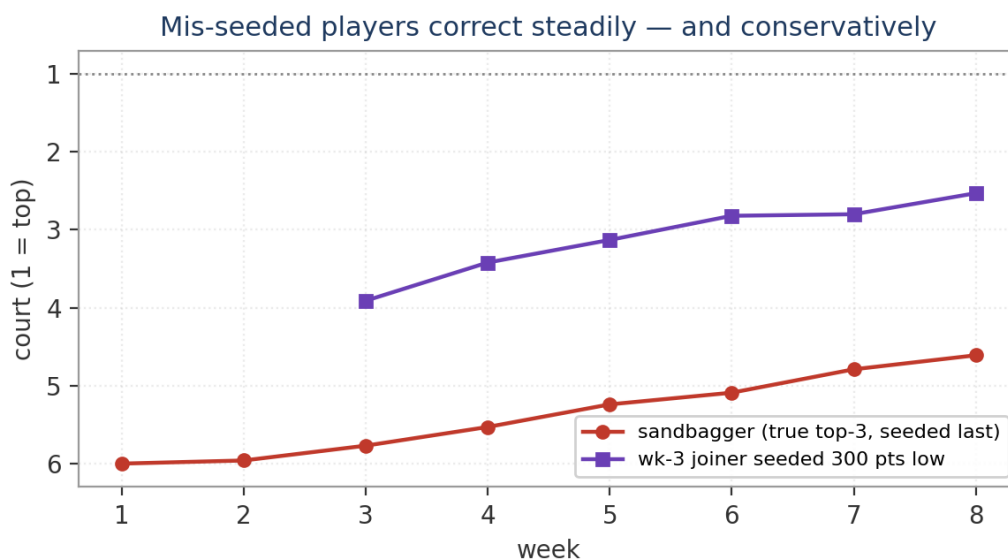


Figure 2. A last-seeded strong player and a 300-point-low mid-season joiner both correct steadily under the production configuration — the joiner reaches the upper courts; the extreme 700-point case deliberately does not finish the trip in eight weeks.

## 5. Operating the System

**Week one.** Sort every player by DUPR or shadow grade from highest to lowest and deal courts — top six to court one, and so on. Ties are common because DUPR moves in coarse steps; players at the same number are genuinely indistinguishable on day one, so list order is fair. Where a tie straddles a court boundary, breaking it in favor of the player with a real DUPR is a reasonable nicety but is not required.

**Each week thereafter.** Provide two things: the game scores from the session, or at minimum who won each game, and the list of who was present. The DUPR seeds are needed only once. The whole session is scored as one batch against the week-start ladder, so score-sheet order never matters. The shadow-seed correction retires itself after roughly two sessions per player; from then on the ladder is driven entirely by ordinary results.

**Absences.** An absent player's rating is frozen and their update factor temporarily enlarged on return. Their displayed position may drift as active players move around them; participant-facing ladders may list active players only.

**Interventions.** Correct a clearly erroneous seed by hand — not a workaround but the designed remedy for an implausible seed, and the first week or two is the natural window. Beyond that the system runs untouched: ranking is a straight sort of ratings, so a court always reflects points, and a large jump appears only when results justify it.

## Appendix A. Production Parameters

The deployed configuration — the settings every result in this paper was measured at. All are exposed as configuration for the organizer. This configuration was introduced in commit f536c62 and this document verified against repository state 5a00c2e (July 2026).

| Parameter                          | Production value                                                        |
|------------------------------------|-------------------------------------------------------------------------|
| Seed anchor                        | 1500 at DUPR 3.5                                                        |
| Seed scale                         | 200 points per DUPR                                                     |
| K (new player)                     | 56                                                                      |
| K (steady state)                   | 32                                                                      |
| K decay                            | 10 games                                                                |
| Win/margin weight                  | 0.5 / 0.5                                                               |
| Contribution share                 | clipped to 35%–65%                                                      |
| Winner floor                       | enabled                                                                 |
| Absence boost                      | +15% per missed week, maximum 1.5×                                      |
| Seed-correction blend (b)          | 0.1                                                                     |
| Seed-correction decay ( $\tau_p$ ) | 16 games                                                                |
| Dead zone (z)                      | 150 points                                                              |
| Initial pull clip                  | $\pm 400$                                                               |
| Pull taper                         | linear to zero over 12 games                                            |
| Pull eligibility                   | shadow grade, <12 prior games, $\geq 3$ -game all-win or all-loss sweep |
| Rank clamp                         | disabled — rank is a strict rating sort                                 |
| Rating floor                       | none — ratings are unbounded internal state                             |

## Appendix B. Notation

Symbols used in the equations of Section 2. Angle brackets denote a session mean — an average taken over all of a player's games in a given week.

| Symbol                                                    | Meaning                                                                                                                      |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| $R_i, R_{\text{team}}$                                    | Rating of player $i$ ; team rating (the mean of the two partners).                                                           |
| $R^{\text{start}}, R^{\text{after}}$                      | A player's rating at the frozen start of a session, and after the batch of accumulated deltas is applied.                    |
| $E_A$                                                     | Elo win expectation for team A against team B.                                                                               |
| $S, W, \beta$                                             | Blended game score in $[0,1]$ ; win indicator (1 win, 0 loss); margin weight (0.5).                                          |
| $s_f, s_a$                                                | Points scored for and against in the game.                                                                                   |
| $K(g)$                                                    | Update factor at $g$ games played, decaying from $K_0 = 56$ to $K_\infty = 32$ .                                             |
| $\tau_K, \tau_p$                                          | Decay constants (in games) for the update factor (10) and the pull strength (16).                                            |
| $\Delta_{\text{team}}, \Delta_i$                          | Team rating change; player $i$ 's renormalized share of it. A winning player's $\Delta$ is floored at zero.                  |
| $w_i$                                                     | Contribution share of player $i$ , clipped to $[0.35, 0.65]$ and renormalized.                                               |
| $m, a$                                                    | Absence multiplier (capped at 1.5); number of consecutive missed weeks.                                                      |
| $\langle O \rangle, \langle T \rangle, \langle S \rangle$ | Session-mean opponent-team rating, partner rating, and team score (clipped to $[0.08, 0.92]$ ).                              |
| $L$                                                       | Session Elo margin: the rating gap implied by the mean score, $400 \log_{10}(\langle S \rangle / (1 - \langle S \rangle))$ . |
| $P$                                                       | Session performance rating (the rating implied by the week's results).                                                       |
| $g_p$                                                     | Games played entering the session; the correction requires $g_p < 12$ .                                                      |
| $\lambda, b, z, c(g_p)$                                   | Pull strength; blend ceiling (0.1); dead zone (150); clip, tapering from $\pm 400$ to zero over 12 games.                    |

## Appendix C. Worked Examples

One game, one short rivalry, and one debut, carried through the arithmetic so the formulas of Section 2 can be seen operating on real numbers. Every figure is produced by the reference engine at the parameters of Appendix A.

### C.1 A single game, recorded and scored

Four settled players take the court. Their current ratings are  $A = 1580$ ,  $B = 1500$  (team 1) against  $C = 1540$ ,  $D = 1520$  (team 2); all four are at the steady-state update factor  $K = 32$ . They play to 11 and team 1 wins 11–9. The organizer records only what happened on the court — who partnered whom, and the score:

| Game | Team 1 | Team 2 | Score  |
|------|--------|--------|--------|
| 1    | A + B  | C + D  | 11 – 9 |

Table C-1. The complete record an organizer enters for a game: partners on each side and the final score.

From that record the engine computes the update. The team ratings are the partner averages,  $R_1 = 1540$  and  $R_2 = 1530$ , so team 1 is expected to win with probability  $E = 0.514$  (equation 4). The 11–9 result blends with the win to a score  $S = 0.795$  (equation 5), a surprise of 0.281, and a team swing of  $32 \times 0.281 = +8.99$  points to team 1, mirrored to team 2. The contribution split then divides each team's swing between its partners by rating (equations 7–8):

| Player | Rating before | Change | Rating after |
|--------|---------------|--------|--------------|
| A      | 1580          | +9.23  | 1589.23      |
| B      | 1500          | +8.76  | 1508.76      |
| C      | 1540          | -9.05  | 1530.95      |
| D      | 1520          | -8.94  | 1511.06      |

Table C-2. The same game after scoring. Within each team the higher-rated partner moves slightly more — A gains more than B, C loses more than D.

Notice two properties. Within team 1, the stronger player gains a little more (9.23 vs 8.76): the presumed carrier is credited more, even in a shared win. And the winners' gains exactly offset the losers' losses — in this game, because all four share the same  $K$  and the floor never binds. When those conditions fail, games are only approximately zero-sum, which is why season drift is measured (+0.045 points per player, Section 4) rather than assumed away.

### C.2 Two rivals over three batch-scored weeks

A toy league of six — evenly matched rivals P and Q plus four supporting players — all seeded 1500 as new players, all verified so the seed correction never applies. Two games per week are shown; the engine treats every game identically. Watch the batch rule: within each week, every game is scored against the ratings and update factors that opened the week, and the deltas land together at the end.

| Week | Recorded game (team vs team, score) | P      | Q      |
|------|-------------------------------------|--------|--------|
| 1    | P+S beat M+N, 11–8                  | +17.82 | —      |
| 1    | Q+M beat P+T, 11–9                  | -16.55 | +16.55 |
| 2    | Q+S beat M+N, 11–7                  | —      | +16.24 |
| 2    | Q+T beat P+N, 11–9                  | -14.74 | +15.40 |
| 3    | P+T beat M+S, 11–9                  | +15.79 | —      |
| 3    | P+M beat Q+N, 11–8                  | +16.92 | -17.44 |

Table C-3. Two games per player per week, with each player's rating change computed against the week-start snapshot. A dash means that player was not in that game.

The batch rule is visible inside week 1: P's win (+17.82) and head-to-head loss to Q (-16.55) are both computed at P's week-start state — 1500 rating,  $K = 56$  — regardless of listed order; reversing the score sheet would change nothing. Q's week-2 pair surges ahead; P answers in week 3 with the head-to-head win that pulls the two back together:

|          | Start | End wk 1 | End wk 2 | End wk 3 |
|----------|-------|----------|----------|----------|
| P rating | 1500  | 1501.27  | 1486.53  | 1519.24  |
| Q rating | 1500  | 1516.55  | 1548.19  | 1530.74  |

Table C-4. The rivalry as ratings. Both start level; three weeks later they sit a dozen points apart — an honest reflection of two players of about the same standard.

Each week these players move a few dozen points at most, though nothing forces the modesty — ratings simply do not swing far enough in one week to jump multiple courts, which is exactly how the ladder stays stable without a movement cap.

### C.3 A shadow debut and the seed correction

Finally, the safety valve. Player X debuts with a shadow grade of DUPR 3.0, seeding at 1400 (equation 1), and has the maximum possible night: four games, four wins,  $K = 56$  throughout the frozen batch:

| Game | X's partner (rating) | Opposing team (avg rating) | Score | X's $\Delta$ (ordinary Elo) |
|------|----------------------|----------------------------|-------|-----------------------------|
| 1    | 1450                 | 1450                       | 11-6  | +21.98                      |
| 2    | 1520                 | 1475                       | 11-8  | +18.24                      |
| 3    | 1480                 | 1480                       | 11-9  | +19.21                      |
| 4    | 1500                 | 1440                       | 11-5  | +20.11                      |

Table C-5. A debut sweep. Ordinary Elo alone awards X +79.55 points, taking the rating to 1479.55.

The sweep qualifies for the seed correction: X is shadow-graded, entered with 0 prior games, played at least three games, and won them all. The session means are  $\langle S \rangle = 0.8409$ ,  $\langle O \rangle = 1461.2$ ,  $\langle T \rangle = 1487.5$ , and the session's Elo margin is  $L = +289.2$  (equation 13), giving a performance rating  $P = 2013.5$  (equation 12) — the day looked like the play of a ~2000-rated player, though on a four-game sample. The gap to the post-Elo rating is 533.9 points; the 150-point dead zone reduces it to 383.9, the debut clip of  $\pm 400$  leaves that intact (equation 15), and the blend  $\lambda = 0.1$  turns it into a pull of +38.39 (equation 16). X finishes the night at 1517.94.

This is the correction working as intended: a maximally impressive debut earns about 118 points, of which at most 40 can ever come from the correction. The same night under the pre-revision 0.6 blend would have added roughly 230 points beyond ordinary Elo — the over-reaction that motivated Section 4.1. A 0-4 sweep would pull the mirror image down. And after two such sessions the correction retires: from game twelve on, X advances like everyone else.

*This document accompanies the reference engine (engine/courtrank.py) and production pipeline (engine/pipeline/ingest.py), which define the deployed configuration and reproduce every figure and table herein. Design history is recorded in docs/decisions/ (ADRs 0001-0005).*